

Unit 6 – Inference for Categorical Data: Proportions

Notes 6 – Relationship between Confidence Intervals and Significance Tests

Comparing Inference Procedures

A recent study suggested that 77% of teenagers have texted while driving. A random sample of 50 teenage drivers at the school was taken and 33 admitted to texting while driving. Assume all conditions have been met. Use your calculator to answer the questions below.

- a) Construct a 99% confidence interval to estimate the true parameter of teens who text while driving.
- b) Conduct a significance test to determine if the proportion of teens who text while driving is different than 77%. Use a 1% significance level.

- c) What type of results did you get for the confidence interval and significance test?

Confidence Intervals 🍷 Significance Testing

$100(1 - \alpha)\%$ confidence interval	$\leftrightarrow$	significance level α with $H_0: p = p_0$ $H_a: p \neq p_0$
90% Confidence Interval	$\leftrightarrow$	_____ Significance Level
_____ Confidence Interval	$\leftrightarrow$	5% Significance Level
99% Confidence Interval	$\leftrightarrow$	_____ Significance Level

- If the confidence interval contains the null hypothesis from a two-sided test, we would
- If the confidence interval does not contain the null hypothesis from a two-sided test, we would

Example

Scott's Turf Lawn Builder makes the claim that, when a user plants their grass seeds, 88% of seeds will germinate. The germination rate of seeds is defined as the proportion of seeds that, when properly planted in the fertilizer and watered, sprout and grow. The company regularly tests this claim, to make sure its product is efficient. They plant 500 grass seeds in their fertilizer and find that 412 of the seeds germinate. Assume that all conditions are met.

a) What is the 95% confidence interval for this observation?

b) Suppose the company conducted a test of $H_0: p = 0.88$ against the alternative $H_a: p \neq 0.88$, using $\alpha = 0.05$. Use the confidence interval to determine whether this test would reject or fail to reject the null hypothesis. Explain your reasoning.

c) Find the p-value for the test. Explain what the p-value measures in the context of the problem.